

RMO(INDIA) – 2012(4)

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
 - All Questions carry equal marks.
 - Answer to each questions starts on a new page. Clearly indicate the question number.
1. Let ABCD be a unit square. Draw a quadrant of a circle with A as centre and B, D as end points of the arc. Similarly draw a quadrant of a circle with B as centre and A, C as end points of the arc. Inscribe a circle Γ touching the arc AC internally, the arc BD internally and also touching the side AB. Find the radius of the circle Γ .
 2. Let a, b and c are positive integers such that a divides b^2 , b divides c^2 and c divides a^2 . Prove that abc divides $(a + b + c)^7$.
 3. Let a and b are positive real numbers such that $a + b = 1$. Prove that $a^a b^b + a^b b^a \leq 1$.
 4. Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$. Find the number of pairs $\{A, B\}$ such that $A \subseteq B, B \subseteq X, A \neq B$ and $A \cap B = \{4, 5, 7, 8, 9, 10\}$.
 5. Let ABC be a triangle. Let E be a point on the segment BC such that $BE = 2EC$. Let F be the mid point of AC. Let BF intersect AE in Q. Determine BQ/QF.
 6. Solve the system of equations for positive real numbers: $\frac{1}{xy} = \frac{x}{z} + 1, \frac{1}{yz} = \frac{y}{x} + 1, \frac{1}{zx} = \frac{z}{y} + 1, .$

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